

DBMS-WEAK 3 PROJECT

INVENTORY MANAGEMENT SYSTEM

The e-commerce system maintains information about sellers and inventory. The system stores seller details and tracks the stock quantity, stock status, and latest update date of products supplied by different sellers. The system helps manage seller information and monitor product inventory to ensure that product stock and availability can be tracked effectively.

1. Creating Seller and Inventory table

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25), Email  
    VARCHAR(30), Phone number(10), Address VARCHAR(45));
```

Table created.

```
SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),  
    2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),  
    3 Stock_Quantity number, Stock_status VARCHAR(11),  
    4 Last_Updated DATE);
```

Table created.

2. Establish Relationships

Using Primary Key Primary keys are used to uniquely identify records in the Seller and Inventory tables.

- Seller_ID is the primary key of the Seller table.
- Inventory_ID is the primary key of the Inventory table.

The Inventory table is connected to both the Product and Seller tables using foreign keys. Seller → Inventory One seller can supply multiple products. Therefore, the relationship between Seller and Inventory is One-to-Many (1:M). Product → Inventory A product can have inventory information associated with it. The Inventory table uses Product_ID to identify the corresponding product.

```
SQL> ALTER TABLE SELLER  
    2 MODIFY EMAIL VARCHAR2(100) NOT NULL;
```

Table altered.

```
SQL> ALTER TABLE SELLER  
    2 ADD CONSTRAINT UQ_SELLER_EMAIL UNIQUE (EMAIL);
```

Table altered.

```
SQL> ALTER TABLE INVENTORY  
    2 ADD CONSTRAINT CHK_STOCK_QUANTITY CHECK (STOCK_QUANTITY > 0);
```

Table altered.

3. Foreign key for inventory table:

Foreign Keys

- Product_ID in the Inventory table references Product_ID in the Product table.
- Seller_ID in the Inventory table references Seller_ID in the Seller table.

```
SQL> ALTER TABLE INVENTORY
  2  ADD CONSTRAINT CHK_STOCK_QUANTITY CHECK (STOCK_QUANTITY > 0);

Table altered.
```

```
SQL> ALTER TABLE INVENTORY
  2  ADD CONSTRAINT FK_INVENTORY_PRODUCT
  3  FOREIGN KEY (PRODUCT_ID)
  4  REFERENCES PRODUCT(PRODUCT_ID);
```

Constraints Unique Email ID A UNIQUE constraint is added to the Email column in the Seller table. This ensures that two sellers cannot have the same email address.

```
SQL> ALTER TABLE SELLER
  2  ADD CONSTRAINT UQ_SELLER_EMAIL UNIQUE (EMAIL);

Table altered.
```

Email NOT NULL: The NOT NULL constraint is applied to the Email column. This ensures that every seller record contains an email address.

```
SQL> ALTER TABLE SELLER
  2  MODIFY EMAIL VARCHAR2(100) NOT NULL;

Table altered.
```

Add Constraint to Check Stock Quantity is Positive: The CHECK constraint ensures that the stock quantity in the Inventory table is always greater than zero.

```
SQL> ALTER TABLE INVENTORY
  2  ADD CONSTRAINT CHK_STOCK_QUANTITY CHECK (STOCK_QUANTITY > 0);

Table altered.
```

Inserting Values into Seller Table

```

QL> INSERT INTO SELLER (SELLER_ID, SELLER_NAME, EMAIL, PHONE, ADDRESS)
  2 VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', '9876543210', 'Chennai');

row created.

QL>
QL> INSERT INTO SELLER (SELLER_ID, SELLER_NAME, EMAIL, PHONE, ADDRESS)
  2 VALUES (2, 'Priya Sharma', 'priya@gmail.com', '9876543211', 'Coimbatore');

row created.

QL>
QL> INSERT INTO SELLER (SELLER_ID, SELLER_NAME, EMAIL, PHONE, ADDRESS)
  2 VALUES (3, 'Arun Kumar', 'arun@gmail.com', '9876543212', 'Madurai');

row created.

QL>
QL> INSERT INTO SELLER (SELLER_ID, SELLER_NAME, EMAIL, PHONE, ADDRESS)
  2 VALUES (4, 'Divya Raj', 'divya@gmail.com', '9876543213', 'Salem');

row created.

```

SELLER_ID	SELLER_NAME	EMAIL	PHONE	ADDRESS
1	Ravi Kumar	ravi@gmail.com	9876543210	Chennai
2	Priya Sharma	priya@gmail.com	9876543211	Coimbatore
3	Arun Kumar	arun@gmail.com	9876543212	Madurai
4	Divya Raj	divya@gmail.com	9876543213	Salem

6.Inserting Values to Inventory table

```

SQL> INSERT INTO INVENTORY
  2 (INVENTORY_ID, PRODUCT_ID, SELLER_ID, STOCK_QUANTITY, STOCK_STATUS, LAST_UPDATED)
  3 VALUES
  4 (1, 101, 1, 25, 'Available', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO INVENTORY
  2 (INVENTORY_ID, PRODUCT_ID, SELLER_ID, STOCK_QUANTITY, STOCK_STATUS, LAST_UPDATED)
  3 VALUES
  4 (2, 102, 2, 40, 'Available', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO INVENTORY
  2 (INVENTORY_ID, PRODUCT_ID, SELLER_ID, STOCK_QUANTITY, STOCK_STATUS, LAST_UPDATED)
  3 VALUES
  4 (3, 103, 3, 15, 'Low Stock', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO INVENTORY
  2 (INVENTORY_ID, PRODUCT_ID, SELLER_ID, STOCK_QUANTITY, STOCK_STATUS, LAST_UPDATED)
  3 VALUES
  4 (4, 104, 4, 30, 'Available', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

```

```
SQL> SELECT*FROM INVENTORY;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATE
1	101	1	25	Available	22-SEP-26
2	102	2	40	Available	22-SEP-26
3	103	3	15	Low Stock	22-SEP-26
4	104	4	30	Available	22-SEP-26

```
SQL> SELECT TABLE_NAME, CONSTRAINT_NAME, CONSTRAINT_TYPE
2 FROM USER_CONSTRAINTS
3 WHERE TABLE_NAME IN ('SELLER', 'INVENTORY', 'PRODUCT');
```

TABLE_NAME	CONSTRAINT_NAME	CONSTRAINT_TYPE
INVENTORY	SYS_C007291	C
INVENTORY	SYS_C007292	C
INVENTORY	SYS_C007293	P
INVENTORY	FK_INVENTORY_SELLER	R
PRODUCT	SYS_C007295	C
PRODUCT	SYS_C007296	P
SELLER	SYS_C007290	P

7 rows selected.

Report 1

Seller-Wise Product Report Purpose This report displays the number of products handled by each seller and the total available stock associated with the seller.

```
SQL> SELECT
2 P.PRODUCT_NAME,
3 I.STOCK_QUANTITY,
4 I.STOCK_STATUS
5 FROM PRODUCT P
6 JOIN INVENTORY I
7 ON P.PRODUCT_ID = I.PRODUCT_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
Laptop	25	Available
Mobile Phone	40	Available
Headphones	15	Low Stock
Keyboard	30	Available

```
SQL> SELECT
  2     S.SELLER_NAME,
  3     COUNT(I.PRODUCT_ID) AS NUMBER_OF_PRODUCTS,
  4     SUM(I.STOCK_QUANTITY) AS TOTAL_STOCK
  5 FROM SELLER S
  6 JOIN INVENTORY I
  7 ON S.SELLER_ID = I.SELLER_ID
  8 GROUP BY S.SELLER_NAME;
```

SELLER_NAME		
NUMBER_OF_PRODUCTS	TOTAL_STOCK	
Ravi Kumar	1	25
Divya Raj	1	30
Priya Sharma	1	40

SELLER_NAME		
NUMBER_OF_PRODUCTS	TOTAL_STOCK	
Arun Kumar	1	15

Report 2

Stock Availability Report Purpose This report displays the product name, stock quantity and current stock

```
SQL> SELECT
  2     P.PRODUCT_NAME,
  3     I.STOCK_QUANTITY,
  4     I.STOCK_STATUS
  5 FROM PRODUCT P
  6 JOIN INVENTORY I
  7 ON P.PRODUCT_ID = I.PRODUCT_ID;
```

PRODUCT_NAME	
STOCK_QUANTITY	
STOCK_STATUS	

Laptop	25
Available	
Mobile Phone	40
Available	

PRODUCT_NAME	
STOCK_QUANTITY	
STOCK_STATUS	

Headphones	15
Low Stock	
Keyboard	30

Report 3

Inventory Analysis The Inventory Analysis report provides information about the overall inventory maintained in the system. Total Available Products The following query is used to count inventory records whose stock status is Available.

```
SQL> SELECT COUNT(*)AS TOTAL_AVAILABLE_PRODUCTS
  2  FROM INVENTORY
  3  WHERE STOCK_STATUS='AVAILABLE';

TOTAL_AVAILABLE_PRODUCTS
-----
                        0
```

Product Out of Stock:

```
SQL> SELECT COUNT(*)AS OUTOFSTOCK_PRODUCTS
  2  FROM INVENTORY
  3  WHERE STOCK_STATUS='OUTOFSTOCKS';

OUTOFSTOCK_PRODUCTS
-----
                        0
```

Average Inventory Quantity:

```
SQL> SELECT AVG(STOCK_QUANTITY)AS AVERAGE_INVENTORY FROM INVENTORY;

AVERAGE_INVENTORY
-----
                27.5
```